INTERNSHIP RESEARCH PROPOSAL

ECONOMIC VIABILITY OF BIO-BRICKS

LIFE CYCLE COSTING (LCC) ANALYSIS

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PROJECT CONTEXT

THE YUCATAN PENINSULA CRISIS

THE SARGASSUM CHALLENGE

SOCIO-ECOLOGICAL IMPACT

Massive brown algae blooms wash ashore annually, damaging marine ecosystems and local tourism.

Decomposition releases toxic gases, creating urgent public health risks for coastal communities (hydrogen sulfide and ammonia).

The MAYA-CO2 project aims to valorize this biomass into functional building materials.

MAYA-CO2 INNOVATION PILLARS

CIRCULAR BIOECONOMY

Converting invasive biomass into a provisioning service, replacing high-carbon masonry units.

CARBON SINKS

Permanently locking biogenic carbon into urban structures to mitigate global warming potential.

AIR PURIFICATION

Integrating bio-based MOFs to actively adsorb indoor air pollutants within the brick structure.

ECONOMIC JUSTIFICATION

60%

INNOVATION FAILURE RATE

BRIDGING THE FINANCIAL GAP

Sustainable innovations often fail due to the "Green Premium" compared to high-carbon incumbents.

Without tiered Life Cycle Costing (LCC), bio-bricks risk pricing themselves out of target markets.

This proposal aims to quantify the cost barrier for commercial scale-up.

RESEARCH TRACK: LCC METHODOLOGY

TECHNICAL COSTING

Analyzing chemical precursors for MOFs, kiln energy requirements in Mexico, and localized labor costs for algae harvesting.

TIERED BENCHMARKING

Simultaneously evaluating bio-brick costs against low-income housing baselines and premium sustainable architecture segments.

MARKET SEGMENTATION

AFFORDABLE HOUSING

Benchmarking against standard Mexican cinder blocks (CMU) to test viability for mass residential projects.

PREMIUM ECO-TOURISM

Benchmarking against Autoclaved Aerated Concrete (AAC) where developers prioritize LEED-certified sustainable materials.

DATA STRATEGY AND LCC COMPONENTS

CATEGORY	SOURCE	LCC COMPONENT (THEORY)
Incumbent Materials	CMIC Price Index	Acquisition Costs
MOF Precursors	Bulk B2B Platforms	Operating Costs (Direct Materials)
Local Labor	CONASAMI Wage Data	Operating Costs (Direct Labor)
Energy Inputs	CFE Industrial Tariffs	Operating Costs (Utilities)

IMPLEMENTATION ROADMAP

WEEK 1

Boundaries

Defining functional units and cradle-to-gate boundaries for LCC modeling.

WEEKS 2-3

Data Search

Gathering material costs and technical specs from Mexican industrial indices (CMIC, CFE).

WEEK 4

Modeling

Building dynamic financial models in Excel to isolate the "Tech Premium" of MOF bricks.

WEEK 5

Testing

Sensitivity analysis on energy price spikes and varying chemical precursor bulk rates.

WEEK 6

Reporting

Synthesis of dashboards and final LCC reporting for project stakeholders.

COST BENCHMARKING (MXN/M²)

**Figures are hypothetical placeholders to be validated during the internship research phase.*

QUESTIONS?

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